AGRIVIVE / KHAL-FLOW

PUNJAB / FIELD-WATERCOURSE / MANUAL SLIDE GATE

SERIES 20 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



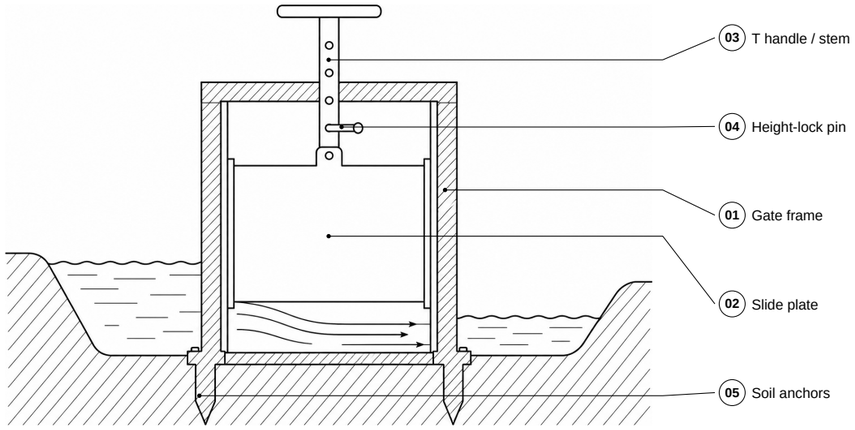
STATE CONTEXT / INDIGENOUS INSPIRATION

Punjab's irrigation watercourses distribute canal water to fields. KHAL-FLOW proposes a small adjustable slide gate for a trial watercourse; gate position alone does not establish volumetric entitlement.

Practice source: Punjab Water Resources Department: e-Sinchai | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 140 W x 70 D x 160 H

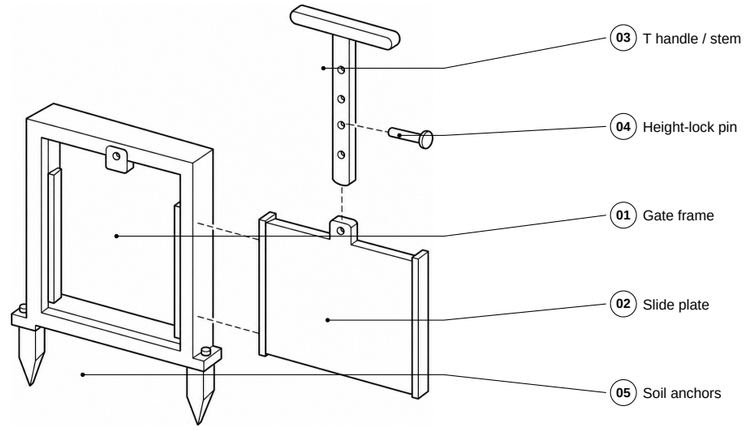


DESIGN / OPERATION NOTES

Seat the frame across a shallow watercourse with a stable downstream path.
Insert the plate in its guides and set height with the retaining pin.
Inspect the opening and clear sediment from the guide channels.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
140 W x 70 D x 160 H

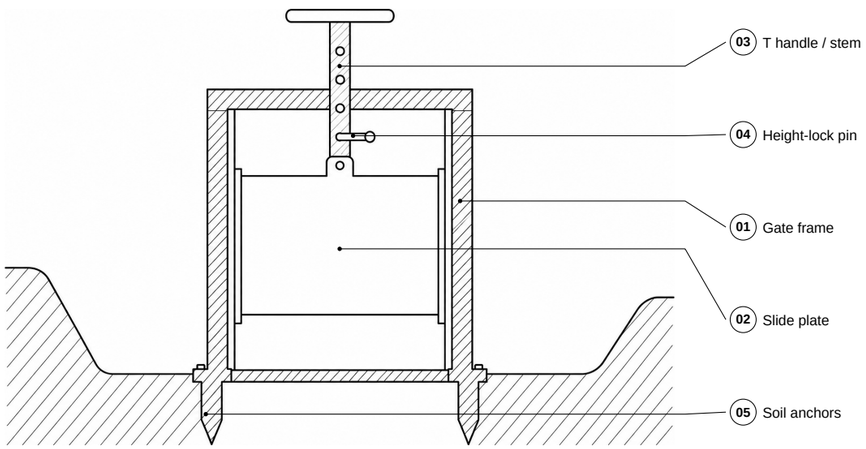


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

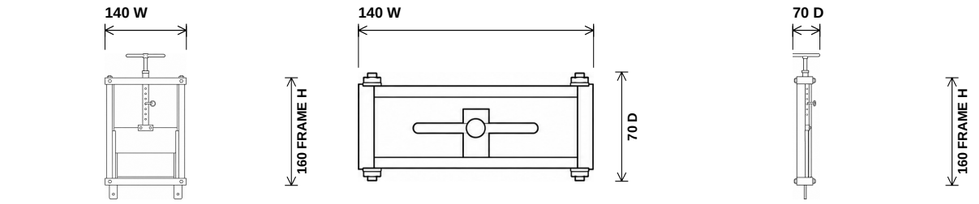
SECTION BASIS / mm
Clear gate window 80 W x 60 H; anchor projection TBD



DESIGN / OPERATION NOTES

Insert the plate in its guides and set height with the retaining pin.
Inspect the opening and clear sediment from the guide channels.

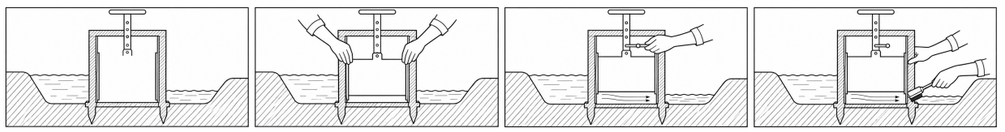
03 ORTHOGRAPHIC STUDIES / mm



FRONT / FRAME **TOP / FRAME** **SIDE / FRAME**

Frame 140 W x 70 D x 160 H, excluding unresolved anchor projection. Clear window 80 W x 60 H.

05 INSTALLATION / OPERATING SEQUENCE



1 SEAT FRAME **2 INSERT GATE** **3 SET / PIN** **4 INSPECT GUIDES**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Frame W x D x H
140 x 70 x 160

Clear window W x H
80 x 60

Anchor projection
TBD

02 GEOMETRIC CHECK

CLEAR GATE WINDOW

4,800.0 mm2

80 x 60
Gate position does not establish fair volumetric delivery

03 PARTS / QTY

01 Gate frame	**1**
02 Slide plate	**1**
03 T handle / stem	**1 set**
04 Height-lock pin	**1**
05 Soil anchors	**2 integral**

Frame/plate: candidate polymer. Handle/pin and sealing materials TBD.

04 INTERFACES / CHECKS

Test leakage, plate friction and pin retention.

Measure stage-discharge behaviour in trials.

Check local scour and anchor projection.

05 DATUMS / DRAWING BASIS

A: frame base. B: guide centres. Establish pin settings from the selected opening.
PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-20 / P01	KHAL-FLOW	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	20 / 28